

MetaQSAR: A comprehensive tool for automated QSAR modeling

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1 Introduction

Quantitative structure–activity relationship (QSAR) modeling has become an indispensable tool for specialists in diverse scientific fields, such as pharmacologists, medicinal chemists, toxicologists, or materials scientists, to predict compound properties, streamline decision-making, and design further experiments effectively (Muratov et al., 2020). QSAR models can be used to design and optimize new bioactive compounds, to assess the impact of chemical compounds on human health and the environment, as well as to elucidate the properties of complex materials, such as nanomaterials, aiding in the design of more effective materials for industrial applications (Li et al., 2022). Despite their broad applicability, researchers in these disciplines often lack expertise in key areas such as data handling, data curation, statistical analysis, machine learning, and/or programming, which are essential for developing robust QSAR models. Moreover, extensive experience is often required to recognize and overcome the challenges associated with QSAR modeling (Scior et al., 2009). This gap highlights the need for user-friendly, automated QSAR platforms to empower scientists in independently building and applying QSAR models for their specific purposes.

In today's digital era, vast amounts of data are continuously generated and made accessible through chemical databases such as ChEMBL (Gaulton et al., 2017), BindingDB (Liu et al., 2025), PubChem (Kim et al., 2016), ECOTOX (Olker et al., 2022), and the OpenFoodTox chemical hazards database (Dorne et al., 2021). To ensure the relevance and accuracy of QSAR models, it is essential to regularly update and refine them using new data, primarily sourced from the aforementioned databases, which are continuously maintained. Such model updates expand the applicability domain of the models and improve their

predictive quality. The process of building a typical QSAR model includes several key steps: data collection, curation, data normalization, descriptor calculation, model development, model validation, applicability domain determination, and finally model implementation. Automated QSAR platforms (Mansouri et al., 2024; Dixon et al., 2016; Bachorz et al., 2024; Mervin et al., 2024; Cox et al., 2013; Kausar and Falcao, 2018; Nantasenamat et al., 2015; Sushko et al., 2011) play a vital role in streamlining these recurring tasks, minimizing human errors, and enhancing overall efficiency, providing significant benefits to users across various scientific disciplines, serving as indispensable tools for advancing research and promoting innovation.

Preconfigured workflows for data preprocessing, model building, validation, and deployment simplify complex processes, reduce errors, and increase acceptance in diverse scientific fields. Standardized automated workflows also ensure reproducibility of the models, which is crucial for scientific validation. Integration of advanced machine learning methods and statistical approaches further enhances the scientific robustness of the models. Moreover, automation can boost interdisciplinary collaborations as it surpasses the technical barriers and also reduces human biases, resulting in scientifically reliable outcomes.

This chapter provides an overview of existing automated QSAR platforms, highlighting their key advantages and limitations. Moreover, it introduces MetaQSAR, a freely accessible platform specifically developed to bridge these gaps by integrating user-friendly features with scientific precision, enabling researchers to efficiently develop and utilize robust QSAR models. This automated QSAR platform with a user-friendly graphical user interface (GUI) available at <https://chemopredictionsuite.com/MetaQSAR> allows users to perform complex tasks of QSAR modeling in a few clicks, accelerating the learning pace for non-experts.

2 Existing QSAR modeling platforms

Several software platforms (listed in Table 33.1) have been developed to streamline and automate the QSAR modeling process. These platforms aim to assist researchers in generating predictive models with minimal manual intervention while maintaining scientific reliability. Each platform offers unique functionalities designed to address specific requirements in QSAR modeling. However, these tools also present limitations that highlight the necessity for further advancements. For example, some platforms lack robust applicability domain analysis, are not fully automated, or require manual intervention for tasks such as model configuration and validation. Others may be constrained by a limited set of descriptors, commercial licensing restrictions, or difficulties in managing sensitive data on cloud-based systems. Table 33.1 summarizes the key features and limitations of several widely used software platforms for automating the QSAR modeling process.

TABLE 33.1 List of well-known automated QSAR platforms along with useful relevant details.

Platform	Availability (type)	Developer/ Source	Key features	Limitations	Link/Reference
AutoQSAR	Commercial (Standalone)	Schrödinger, Inc.	Supports MLR, PLS, Naïve Bayes, decision trees; automatic model building; descriptor + fingerprint-based	Limited descriptors (497); few feature selection options	Reference (Dixon et al., 2016) https://doi.org/10.4155/fmc-2016-0093
DeepAutoQSAR	Commercial (Standalone)	Schrödinger, Inc.	Deep learning-based automation; high prediction for large datasets	Black-box nature	Website: https://www.schrodinger.com/platform/products/deepautoqsar/
QSPRmodeler	Open-source (Jupyter Notebook)	GitHub (Python-based)	End-to-end QSAR: data processing, machine learning, Hyperparameter optimization using Hyperopt, model validation, model storage	No applicability domain (AD) analysis	Reference (Bachorz et al., 2024) Github page: https://github.com/rafalbachorz/qsprmodeler
QSARtuna	Open-source (Jupyter Notebook)	AstraZeneca	End-to-end QSAR; machine learning and deep learning support; multiple descriptor and fingerprint options; uncertainty estimation and model explainability features using SHapley Additive exPlanations (SHAP) analysis and ChemProp Interpret	Lacks AD analysis	Reference (Mervin et al., 2024) Github page: https://github.com/MolecularAI/QSARtuna
QSAR Workbench (Pipeline Pilot)	Commercial (Workflow)	Dassault Systèmes	Comprehensive end-to-end QSAR automation; includes AD determination; user-friendly	Limited to Pipeline Pilot (Warr, 2012) users	Reference (Cox et al., 2013) Website: https://www.3ds.com/products/biovia/pipeline-pilot

Continued

TABLE 33.1 List of well-known automated QSAR platforms along with useful relevant details—cont'd

Platform	Availability (type)	Developer/Source	Key features	Limitations	Link/Reference
QSAR Framework (KNIME)	Open-source (KNIME Workflow)	GitHub	Full pipeline in modular format: preprocessing, descriptor calculation, feature selection, model building, validation, AD analysis and visualization	Learning curve for workflow design	Reference (Kausar and Falcao, 2018) Github page: https://github.com/Saminakausar/Automated-framework-for-QSAR-model-building
Auto-WEKA	Open-source (Standalone)	Waikato	Automates machine learning algorithm and hyperparameter selection; several feature selection and machine learning techniques	Not QSAR-specific; lacks data preparation, AD, and dataset splitting option	Reference (Nantasenamat et al., 2015) Website: https://ml.cms.waikato.ac.nz/weka
OCHEM	Free (Web-based)	https://ochem.eu	Web platform with integrated database and model building; all essential stages of predictive modeling; supports AD; useful for public datasets	Full automation option not available; no train/test split; less suited for private or sensitive data	Reference (Sushko et al., 2011) Website: https://ochem.eu/home/show.do
Open OCHEM	Open-source (Python)	GitHub	Installable version of OCHEM; allows private deployment and analysis	Full automation option is not available	Github page: https://github.com/openochem/openochem

TABLE 33.2 Diverse range of algorithms available in MetaQSAR software platform.

Modeling step	Algorithms/Techniques
Chemical structure curation	Remove salt, detect aromaticity, and standardize nitro group
Descriptor calculation	[Total 4109 molecular features] Calculates 1D, 2D molecular descriptors and several fingerprints (uses PaDEL-descriptors library (Yap, 2011))
Data Preprocessing	Constant and intercorrelated descriptors are removed based on variance cut-off and correlation coefficient (R^2) cut-off values, respectively.
Dataset division	Random approach and Rational approaches (Kennard-Stone, Euclidean distance-based, activity-sorted)
Feature selection techniques	Genetic algorithm, stepwise selection, best subset selection, Random forest, CfsSubsetEval, InfoGainAttributeEval, GainRatioAttributeEval, CorrelationAttributeEval
Machine learning	Multiple linear regression, partial least square, linear discriminant analysis, Random Forest, support vector machine
Model validation	Internal validation (performed using the training set), cross-validation ^a (leave-one-out method or k-fold applied on the training set) and external validation (conducted using the independent test set)
Applicability domain	Standardization approach, confidence estimation approach

^aNote that the leave-many-out method is not applied in automatic QSAR mode, but it is available in the manual mode.

3 Introducing MetaQSAR: A comprehensive QSAR modeling solution

MetaQSAR is an advanced standalone software designed to streamline QSAR model development with minimal user intervention. MetaQSAR aims to simplify the QSAR modeling process while adhering to rigorous computational standards. It supports both fully automated and manual QSAR workflows, making it suitable for users of all expertise levels. The software follows the good modeling practices recommended in the Organisation for Economic Cooperation and Development (OECD) guidelines (Gissi et al., 2024) and European Chemical Agency (ECHA) regulations (García-Fernández, 2020), ensuring compliance with regulatory applications. This regulatory alignment

makes it a suitable choice for QSAR studies in toxicology, drug discovery, and environmental risk assessment.

A key strength of MetaQSAR is its user-friendly interface, which automates all major steps in QSAR modeling, including basic chemical curation, molecular features calculation, data preprocessing, dataset division, feature selection, model training, validation, and applicability domain determination. The chemical structure curation tasks, such as salt removal, aromaticity detection, and nitro group standardization, as well as the calculation of molecular features (including descriptors and fingerprints), are carried out using the PaDEL-descriptors library (Yap, 2011). Further, the software integrates *in-house* written codes for various feature selection and machine learning algorithms alongside several algorithms that were directly utilized from the Weka library (Frank et al., 2010), offering a diverse range of feature selection and machine learning techniques (listed in Table 33.2). MetaQSAR employs multiple validation strategies to ensure the robustness and generalizability of the developed models. These include internal validation using the training set, cross-validation through the leave-one-out or k-fold method within the training set, and external validation using an independent test set. Each of these validation strategies serves to assess different aspects of model performance, from fitting accuracy to predictive reliability on unseen data. It is important to note that the leave-many-out cross-validation approach is not utilized in the automatic QSAR mode to minimize computational time; however, it remains available in the manual mode for users seeking more rigorous validation. To support thorough model evaluation, MetaQSAR provides a comprehensive set of validation metrics (Roy and Mitra, 2011; Gramatica and Sangion, 2016) for both regression and classification models (as detailed in Table 33.3).

To facilitate the identification of optimal models, MetaQSAR incorporates an internal database of numerous preconfigured model setups, encompassing

TABLE 33.3 Internal and external validations metrics provided by MetaQSAR software.

Model type	Validation type	Validation metrics
Regression models	Internal validation	R^2 , R_a^2 , $Q^2_{(LOO)}$, Standard Deviation of the Error of Prediction (SDEP) _{LOO} , r_m^2 (train) metrics, mean absolute error (MAE), Standard Error of Estimation (SEE)
	External validation	Q^2_{f1} , Q^2_{f2} , Q^2_{f3} , r_m^2 (test) metrics, CCC, MAE
Classification models	Internal/External validation	Mathews correlation coefficient (MCC), Sensitivity, Specificity, Precision, False-Positive rate, False-Negative rate, f-value, Receiver Operating Characteristics-Area Under Curve (ROC-AUC)

a diverse range of hyperparameter values. Based on user-defined inputs such as the allotted modeling time or the specified number of model configurations to evaluate, the software systematically explores these configurations. It then selects the most suitable model based on the key validation metrics calculated on the training set, such as the Mean Absolute Error (*MAE*) or the leave-one-out cross-validated coefficient of determination (Q_{LOO}^2) for regression models and the Mathews correlation coefficient (*MCC*) for classification models. This automated approach streamlines model selection while ensuring robust performance and reliability.

Finally, MetaQSAR includes powerful visualization tools, such as response plots, residual analysis, and receiver operating characteristics (*ROC*) curves, enabling users to interpret model predictions effectively and refine their QSAR models with greater precision. In summary, MetaQSAR provides robust feature selection, comprehensive validation, regulatory compliance, and intuitive interpretability.

3.1 Overview of MetaQSAR automation workflow

MetaQSAR software simplifies QSAR model generation for predicting molecular properties with minimal user intervention. Its intuitive GUI supports both automatic and manual QSAR modeling modes, allowing users regardless of modeling expertise to create robust regression and classification models. Fig. 33.1 shows a screenshot of the MetaQSAR software.

The platform successfully achieves several important objectives:

1. **Computational Workflow Integration:** MetaQSAR integrates the data preprocessing, model building, and validation phases of QSAR models, adhering to the standard practices followed by the QSAR community. The workflow as shown in Fig. 33.2 encompasses:
 - a. Input chemical data processing
 - b. Molecular features calculations
 - c. Data preprocessing (removing noninformative features)
 - d. Dataset division
 - e. Model training using several feature selection and machine learning techniques
 - f. Model validation, including robustness testing (such as Y-randomization test)
 - g. Applicability domain determination
1. **Optimization and Search Metaheuristics:** The software incorporates advanced optimization techniques to efficiently explore molecular parameters and identify optimal model configurations. A robust knowledge base of numerous predefined model configurations enhances the likelihood of achieving high-quality models with strong predictive power. Notably, MetaQSAR employs a powerful feature selection method, i.e., genetic

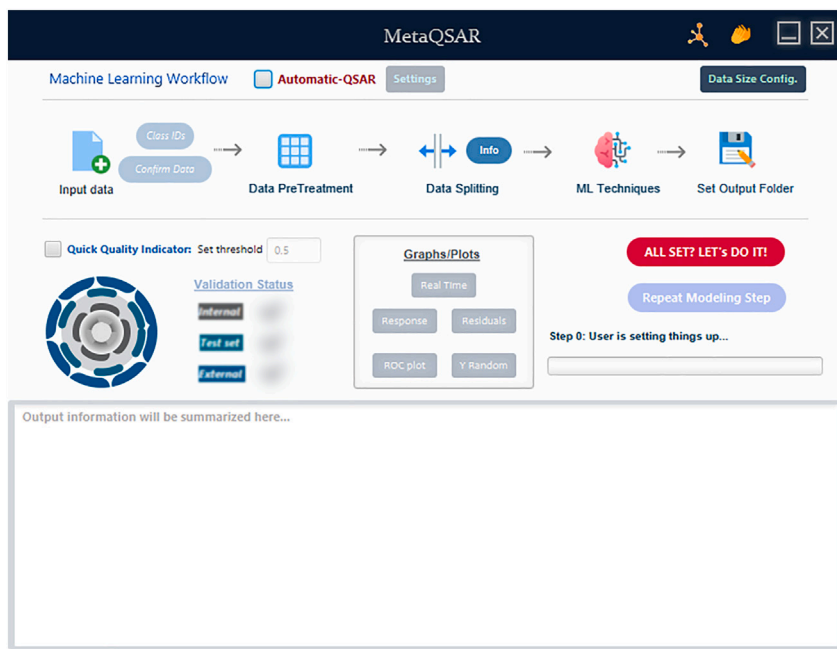
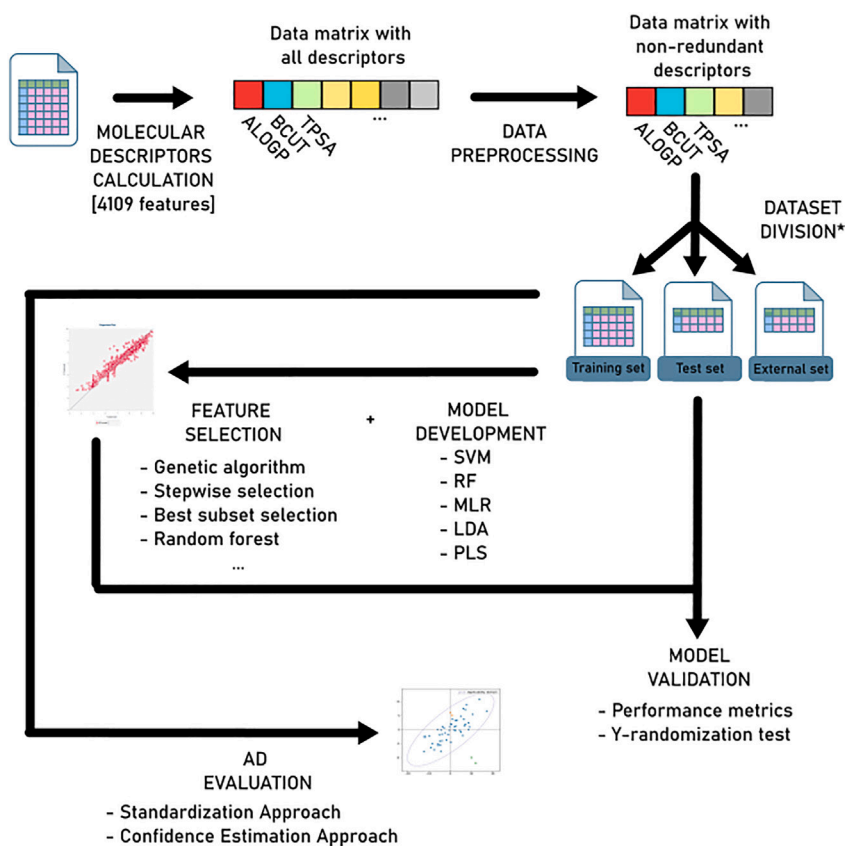


FIG. 33.1 Screenshot of the MetaQSAR software displaying its graphical user interface.

algorithm (GA), among several available techniques. GA uses biologically inspired operators like mutation, crossover, and selection to optimize solutions and identify the most relevant features. Additional feature selection methods include stepwise selection, best subset selection, and random forest, along with several techniques from the Weka library such as CfsSubsetEval, InfoGainAttributeEval, GainRatioAttributeEval, and CorrelationAttributeEval, among others.

2. **Advanced Machine Learning Techniques:** MetaQSAR integrates modern machine learning techniques such as support vector machines (SVM) and random forest (RF), alongside traditional statistical methods like multiple linear regression (MLR), linear discriminant analysis (LDA), and partial least squares (PLS).
3. **Appropriate Model Validation:** In the MetaQSAR platform, regression and classification models undergo rigorous validation using internal (including cross-validation) and external metrics. For small datasets (<50 data points, by default), models are validated solely through cross-validation. Moderate-sized datasets (50–500 data points) are split into training and test sets, with validation performed on the test set. Large datasets (>500 data points) are divided into training, test, and external sets, adding an extra validation layer.



* Dataset division will depend on the number of data points in the initial dataset:

- small datasets (<50 data points): no split
- medium datasets (50-500 datapoints): training set + test set
- large datasets (>500 data points): training set + test set + external set

FIG. 33.2 Workflow of MetaQSAR illustrating automated data preprocessing, model building, and validation steps for QSAR development.

4. Applicability Domain (AD) Assessment: MetaQSAR incorporates AD assessment to evaluate the reliability of model predictions, ensuring that generated QSAR models adhere to regulatory and scientific standards. The software employs two distinct approaches for AD determination: the standardization approach (Roy et al., 2015) and the confidence estimation approach (Mathea et al., 2016), each catering to different modeling scenarios.

4.1 Standardization Approach: This method identifies structural outliers or compounds that fall outside the applicability domain by applying

standardization techniques (Roy et al., 2015). It is compatible with both regression and classification models. By detecting data points significantly deviating from the training set, this approach helps maintain model reliability.

4.2 Confidence Estimation Approach: Designed specifically for classification models, this technique (Mathea et al., 2016) evaluates prediction reliability based on posterior probabilities. A user-defined threshold (default: 0.5) determines whether a classification result is considered reliable. A higher difference between posterior probabilities indicates greater confidence in the prediction. By integrating these approaches, MetaQSAR ensures robust model evaluation, enhancing confidence in QSAR predictions for both classification and regression tasks.

5. Visualizing the results: The output results comprise response plots, residual analysis, Y-randomization test plot, ROC curves, and real-time plot, allowing users to effectively visualize and interpret model predictions. The real-time plot enables users to continuously monitor and visualize the prediction quality of all models generated by the software as they are created. All other visual plots provide valuable insights, enabling users to assess model performance and identify potential areas for improvement. In addition, they assist users in refining their QSAR models with greater precision when utilizing the manual mode.

3.2 MetaQSAR software package and folder structure

The MetaQSAR software package is distributed as a compressed folder containing all essential components required for its operation. Upon extraction, the folder includes several subdirectories and files structured to facilitate a seamless user experience and efficient data management.

- The *Data* folder serves as a repository for sample input files provided with the software, allowing users to familiarize themselves with the input format. In addition, users may store their own input files in this directory for convenient access.
- The *Lib* folder contains all the necessary libraries and supporting files required for the proper functioning of the software. It is imperative to retain this folder in its original location, as any modifications or relocation may impact the software's execution.
- The *Output* folder is designated for storing generated output files, enabling users to systematically organize and retrieve results. While users may specify alternative storage locations, maintaining output files in this dedicated folder ensures streamlined workflow management.

The package's core component is the executable JAR file (*MetaQSAR-v1.0.0.jar*), which serves as the primary launcher for the software. To run the program, users simply need to extract the compressed folder and execute the JAR file; no installation is required. However, Java must be installed on the

system, preferably Java 8, as it includes both Java and JavaFX libraries necessary for proper functionality.

3.2.1 Preparing input files

Preparing input files for MetaQSAR is straightforward, as the software supports comma-separated values (CSV) files with a simple, structured format. Each input file must include three mandatory columns: *chemical ID or serial number*, *SMILES notation*, and *response (activity/toxicity/property) values or class*, all of which can be user-defined. While column headers are flexible, the recommended order chemical ID, SMILES, and response values/class ensures optimal compatibility. To facilitate the process, sample input files are provided in the *Data* folder, serving as a reference for proper formatting.

3.2.2 Output information

MetaQSAR systematically organizes its output into two categories, each corresponding to a distinct stage in the QSAR modeling workflow:

1. Category A: Data preprocessing

- *Descriptor files*: Consist of two files one containing all the calculated molecular features (descriptors and fingerprints), and the other comprising curated features, obtained after removing those with missing or invalid values.
- *Data pretreatment*: Includes intermediate files reflecting curated descriptor sets after successive preprocessing steps, i.e., after removing constant and intercorrelated descriptors.
- *Dataset division files*: Provides training, test, and external set partitions based on the utilized data-splitting method.

2. Category B: Best model obtained so far

- *Model information and Validation results*: Stores QSAR model equations, such as those derived from MLR, and reports all requisite internal and external validation metrics along with the relevant statistical measures for validating both regression and classification-based QSAR models.
- *Model training and test sets files*: These files include compound IDs, selected descriptors, experimental and predicted response values. It also includes posterior probabilities in the case of classification models.
- *Descriptor importance*: Lists ranked descriptors for Random Forest models based on average impurity decrease per attribute over the trees.
- *AD status*: Summarizes results of AD assessment using standardization and confidence estimation approaches.
- *ROC data*: Provides false-positive and true-positive rates for cross-validation sets, enabling custom ROC plot reproduction.
- *Y-randomization test*: Documents results of the Y-randomization test for evaluating model robustness in both regression and classification tasks.

This structured output ensures transparency, facilitating thorough analysis and validation of QSAR models.

3.3 Dual mode functionality

MetaQSAR streamlines QSAR model development through two distinct operational modes: **automatic-QSAR mode** and **manual mode**. These modes accommodate users with varying levels of expertise, offering both automation for ease of use and advanced customization for greater control over model generation.

3.3.1 Automatic-QSAR mode

Designed for users with minimal expertise in computational modeling, the automatic-QSAR mode fully automates the QSAR modeling process. This mode requires minimal user input, involving only the following steps:

- Loading and verifying the input datasets (see Fig. 33.3).
- Configuring basic automatic-QSAR parameters, such as setting the modeling time (in hours) or specifying the number of model configurations to evaluate; additional parameters (as shown in Fig. 33.4) can also be adjusted or fine-tuned based on the initial results generated by the automatic-QSAR mode.
- Specifying the output directory for saving results.

Confirm/Edit Input Data

Input Data information

Number of datapoints = 364

ID column :

CAS_NO

SMILES column :

Canonical Smiles

Response column :

pLD50

☒ Numerical Response

☐ Categorical Response

Confirm and Save

FIG. 33.3 Screenshot of the “Confirm/Edit Input Data” window used for input data verification.

Automatic QSAR Modeling Settings

☐ Set maximum modeling time (in hours) 1

OR

☐ Number of model configurations 10
(maximum possible value 280)

Additional settings:

☒ Selection of modeling type: Both Linear and Non-Linear

Number of features in the model: 3 to 6
(range from minimum to maximum)

Save

FIG. 33.4 Screenshot of the “Automatic QSAR Modeling Settings” window used for configuring basic automatic-QSAR relevant parameters.

That’s all the user needs to do—the software takes care of all essential tasks automatically, including descriptor calculation, data preprocessing, dataset partitioning based on size, feature selection, model training, validation, and applicability domain assessment. By strictly following OECD principles, this workflow ensures scientific rigor and regulatory compliance.

3.3.2 Manual mode

While MetaQSAR’s automatic-QSAR mode efficiently handles the entire modeling process, manual intervention can facilitate more precise fine-tuning. For experienced users seeking enhanced control, the manual mode provides requisite customization options throughout the modeling workflow.

- **Data Preprocessing:** Users can normalize data and remove non-informative descriptors to enhance dataset quality, as shown in [Fig. 33.5](#).
- **Dataset Partitioning:** Multiple approaches are available for splitting data into training and test sets, including random division and rational methods like Kennard-Stone, Euclidean distance-based, and Response-Sorted techniques (see [Fig. 33.6](#)).
- **Feature Selection:** Several variable selection techniques, such as GA, stepwise selection, best subset selection, random forest, CfsSubsetEval, InfoGainAttributeEval, GainRatioAttributeEval, and CorrelationAttributeEval

Data Pre-Treatment: Settings

☐ **Normalize/Scale the features matrix**
(to scale the features values between 0 to 1)

☒ **Remove non-informative descriptors**

Variance cut-off
(to remove constant and near-constant features)

0.001

Inter-correlation R^2 cutoff
(to remove inter-correlated features)

0.95

Save Cancel Reset Default

FIG. 33.5 Screenshot of the “Data Pre-Treatment Settings” window, providing options to normalize data and remove noninformative descriptors.

Dataset Splitting

☐ No Splitting ☒ Split into Two Sets ☐ Split into Three Sets

Select the Dataset Division Method: Random (*Faster)

Seed Number (For Random Division):

Set the Splitting Ratio

Modeling set: 100.0 %

Training set: 70.0 %

Test set: 30.0 %

(True) External set: 0.0 %

Save Cancel Reset Default

FIG. 33.6 Screenshot of the “Dataset Splitting” window, providing multiple options for dividing the datasets.

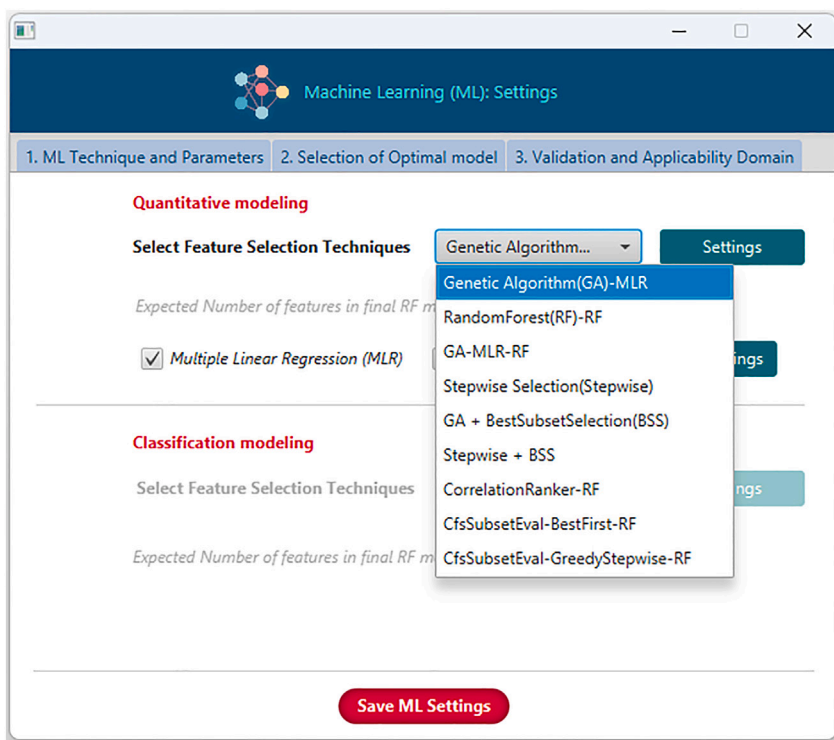


FIG. 33.7 Screenshot of the “Machine learning (ML): Settings” window, providing combinations of several variable selection and machine learning techniques.

are available for manual selection that allow users to identify key descriptors for model building (see Fig. 33.7).

- **Model Development:** Users can choose between linear models (e.g., MLR, LDA) and nonlinear techniques (e.g., RF, SVM) to best fit their dataset and research objectives (see Fig. 33.7).
- **Model Validation:** Comprehensive validation metrics including R^2 , Q^2 , CCC, and MAE are calculated to assess model performance and robustness for both regression and classification models. In Manual Mode, users can also choose the type of cross-validation to apply, such as leave-one-out or leave-many-out (see Fig. 33.8). It also offers options to select the applicability domain determination method, perform additional robustness tests (e.g., Y-randomization), and generate visualization plots such as response plots, residual plots, and ROC curves (see Fig. 33.9).

By providing both automated and customizable workflows, MetaQSAR ensures flexibility and precision in QSAR model generation, making it suitable for both novice and expert users.

1. ML Technique and Parameters 2. Selection of Optimal model 3. Validation and Applicability Domain

Model Validation

Select the Cross-Validation technique:
(Internal Validation)

☒ Leave-One-Out(LOO)

☐ Leave-[M]any-Out, where M = 3

Note: All the recommended internal and external (if applicable) validation metrics will be calculated

Applicability Domain

☒ Standardization Approach

☐ Confidence Estimation Approach, where Threshold = 0.3

Additional Tests, Charts and Plots

☒ Response Plot (Y Obs. vs. Y Pred.)

☒ Residuals Plot

☐ Receiver operating characteristic (ROC) curve

☒ Y Randomization test

Using k-Fold cross validation, K= 10

Generate 'N' random models, N= 50

Save ML Settings

FIG. 33.8 Screenshot of the “Validation and Applicability Domain” window, providing combinations of several variable selection and machine learning techniques.

4 Case studies performed to demonstrate MetaQSAR software performance

To assess the performance and reliability of the MetaQSAR software platform, a series of case studies was conducted using datasets obtained from previously published QSAR studies, mostly focused on various environmental toxicity endpoints. These studies included models developed either by QSAR experts or using other automated QSAR modeling platforms.

In each case, the same dataset used in the original study was employed to ensure fair comparison. MetaQSAR models were developed using its automated pipeline, incorporating feature selection, algorithm selection, model validation, and applicability domain assessment. The models were then evaluated using standard validation metrics for regression and classification, such as R^2 , Q_{LOO}^2 , $RMSE$, and MAE , and *accuracy*, *precision*, *MCC*, *sensitivity*, *specificity*, *f1-score*, and *ROC-AUC*.

4.1 Case study 1: QSAR modeling of personal care products

In this case study, we replicated the QSAR modeling of personal care products (PCPs) for three aquatic organisms—*Pseudokirchneriella subcapitata*, *Daphnia magna*, and *Pimephales promelas*—as originally conducted and reported by



FIG. 33.9 Representative screenshots showcasing various visualization plots generated by the MetaQSAR software.

Gramatica et al. (2016). To ensure consistency and comparability, we employed the same datasets used by the original authors.

For *P. subcapitata*, the dataset comprised 20 PCPs with consistent 96 h growth rate inhibition data (EC_{50} 96 h), in accordance with OECD Test Guideline 201.

In the case of *Daphnia magna*, data on 48 h immobilization (EC_{50} 48 h) were obtained from the ECOTOX database and other sources, as done by the authors

of this study (Gramatica et al., 2016), resulting in a final dataset of 72 PCPs with homogeneous data aligned with OECD Test Guideline 202.

For *P. promelas*, the dataset included 67 PCPs with consistent 96-h mortality data (LC₅₀ 96 h), compiled from the ECOTOX database following OECD Test Guideline 203.

The curated datasets were subsequently employed for QSAR modeling using the automatic QSAR module of the MetaQSAR software platform, and the resulting models were evaluated to compare their performance with that of the previously reported model. All internal and external validation results for the three QSAR models developed by MetaQSAR and reported by Paola Gramatica et al. are presented in the Table 33.4. A comparison of validation metrics shows that the models developed using MetaQSAR are comparable to those created by expert QSAR practitioners. It is important to note that the dataset for *P. subcapitata* contains only 20 data points, classifying it as a small dataset in MetaQSAR. As a result, no training-test split was performed, and model evaluation was carried out solely using cross-validation metrics. The original authors followed a similar approach in modeling this dataset.

4.2 Case study 2: QSAR modeling of soil ecotoxicity in *Folsomia candida*

In this case study, we developed a QSAR model using the MetaQSAR platform to predict the soil ecotoxicity of 54 organic compounds, expressed as 28-day No Observed Effect Concentration (*NOEC*) values in the springtail *Folsomia candida*. The dataset used in our modeling was identical to that employed in a previously published study (Lavado et al., 2022), where the authors had also developed a predictive model based on the same data.

Here, the experimental toxicity data, reported as 28 day *NOECs* (mg/kg dry soil) in accordance with OECD Test Guideline 232, were originally compiled from the EFSA's OpenFoodTox database and the U.S. EPA's ECOTOX database. These values were subsequently converted to a negative decimal logarithmic scale (*pNOEC*), where lower *pNOEC* values indicate lower toxicity.

Using this dataset, we built a predictive QSAR model employing the MetaQSAR software platform and analyzed its performance in comparison to the previously reported model. Notably, in the referenced study, although the dataset contains only 54 data points, the authors split it into a training set (44 data points) and a test set (10 data points). In contrast, MetaQSAR classified it as a small dataset and, therefore, did not perform a training-test split. Instead, the model was evaluated using leave-one-out cross-validation (Q^2_{LOO}). A comparison of the Q^2_{LOO} values (see Table 33.4) indicates that the model developed using MetaQSAR performed slightly better than the one created by expert QSAR practitioners.

TABLE 33.4 Comparison of validation metrics: Reported QSAR models versus MetaQSAR models.

Case study	Endpoint	Model type (machine learning technique)	Validation	Parameters	Already reported model	MetaQSAR model
1	Toxicity to <i>Pseudokirchneriella subcapitata</i>	Regression	Internal	<i>Model (NDesc^c)</i>	<i>MLR^d (Cherkasov et al., 2014)</i>	<i>MLR (Cherkasov et al., 2014)</i>
				<i>Desc. Names</i>	<i>GGI8, Mp</i>	<i>GATS6s, AATSC3m</i>
				<i>NTrain</i>	20	20
				<i>R²</i>	0.93	0.89
				<i>Q_{LOO}²</i>	0.91	0.87
	Toxicity to <i>Daphnia magna</i>	Regression	Internal	<i>Model (NDesc)</i>	<i>MLR (Roy, 2017)</i>	<i>MLR (Martin and Coats, 1983)</i>
				<i>Desc. Names</i>	<i>MW, ATSC0c, GATS1p</i>	<i>ATS1m, SubFPC16, MAXDN, AD2D392</i>
				<i>NTrain</i>	49	51
				<i>R²</i>	0.91	0.88
				<i>Q_{LOO}²</i>	0.89	0.86
			External	<i>NTest</i>	23	21
				<i>Q²_{(test)fin range}</i>	0.80–0.88	0.839–0.84

Continued

TABLE 33.4 Comparison of validation metrics: Reported QSAR models versus MetaQSAR models—cont'd

Case study	Endpoint	Model type (machine learning technique)	Validation	Parameters	Already reported model	MetaQSAR model
	<i>Toxicity to Pimephales promelas</i>	Regression	Internal	<i>Model (NDesc)</i>	<i>MLR (Advances in QSAR Modeling, 2017)</i>	<i>PLS (Martin and Coats, 1983)</i>
				<i>Desc. Names</i>	<i>XLogP, minHother, AVP-7</i>	<i>CrippenLogP, minHother, AVP-7, AATSC2e</i>
				<i>NTrain</i>	46	47
				R^2	0.84	0.90
				Q^2_{LOO}	0.81	0.88
			External	<i>NTest</i>	21	20
				$Q^2_{\text{(test)fn range}}$	0.72–0.81	0.687–0.721
2	<i>Earthworm (Folsomia candida) toxicity</i>	Regression	Internal	<i>Model (NDesc)</i>	<i>PLS^c (8)</i>	<i>MLR (Lombardo et al., 2022)</i>
				<i>NTrain</i>	44	54
				R^2	0.73	0.78
				Q^2_{LOO}	0.64	0.73
			External	<i>NTest</i>	10	<i>NP</i>^b
				$Q^2_{\text{(test)if1}}$	0.68	<i>NP</i>
				$Q^2_{\text{(test)if2}}$	0.64	<i>NP</i>

3	Activity against Pregnane X receptor (PXR)	Classification	Internal	Model (NDesc)	XGBoost (Cronin and Schultz, 2003)	RF ^f (Ojha et al., 2011)
				NTrain	819	819
				Accuracy (Train)	NA ^a	85.23
				Precision (Train)	NA	85.3
				MCC (Train)	NA	70.4
				Sensitivity (Train)	NA	86.02
				Specificity (Train)	NA	84.55
				F1-score (Train)	NA	85.2
				ROC-AUC (Train)	NA	92.6
			External	NTest	91	91
				Accuracy (Test)	82.4	85.7
				Precision (Test)	82.9	85.3
				MCC (Test)	64.8	71.3
				Sensitivity (Test)	82.9	83.72
				Specificity (Test)	80	87.5
				F1-score (Test)	82.9	85.2
				ROC-AUC (Test)	82.4	89.8
4	Toxicity to earthworm (<i>Eisenia foetida</i>) model		Internal	Model (NDesc)	LDA ^g (Cronin and Schultz, 2003)	LDA (Belfield et al., 2023)

Continued

TABLE 33.4 Comparison of validation metrics: Reported QSAR models versus MetaQSAR models—cont'd

Case study	Endpoint	Model type (machine learning technique)	Validation	Parameters	Already reported model	MetaQSAR model
				NTrain	117	115
				<i>Accuracy (Train)</i>	83.76	87.83
				<i>Precision (Train)</i>	70.59	88.4
				<i>Sensitivity (Train)</i>	90.00	87.8
				<i>Specificity (Train)</i>	80.52	77.2
				<i>F1-score (Train)</i>	79.12	87.2
				<i>ROC-AUC (Train)</i>	0.94	0.90
			External	NTest	46	48
				<i>Accuracy (Test)</i>	71.74	72.92
				<i>Precision (Test)</i>	62.50	75.1
				<i>Sensitivity (Test)</i>	58.82	72.9
				<i>Specificity (Test)</i>	79.31	64.9
				<i>F1-score (Test)</i>	60.61	70.9
				<i>ROC-AUC (Test)</i>	0.79	0.75

^aNA, not available

^bNP, not performed

^cNDesc, number of model descriptors

^dMLR, multiple linear regression

^ePLS, partial least squares

^fRF, random forest

^gLDA, linear discriminant analysis

4.3 Case study 3: Classification-based QSAR modeling of pregnane X receptor (PXR) activation

In this case study, we reproduced the classification-based QSAR modeling of the activation effects of the pregnane X receptor (PXR), originally conducted using the QSPRmodeler platform, another automated QSAR development tool discussed earlier in this chapter (Bachorz et al., 2024). Notably, the authors conducted QSAR modeling for both the inhibitory effects on the human androgen receptor (AR) and the activation effects on the PXR. However, we were only able to retrieve the dataset for the PXR receptor, as the dataset for the human AR was unavailable. In the referenced study, the authors retrieved EC₅₀ values for PXR activation from the ChEMBL 3.3 database. The classification model was trained on binarized data, where each molecule was categorized as either *active* or *inactive* based on a threshold EC₅₀ value of 12,000 nM for PXR activity. The curated dataset was then employed by the authors to develop binary classification models using QSPRmodeler. To evaluate and compare modeling performance with QSPRmodeler, we utilized the same dataset and the same threshold to build classification-based QSAR models using our MetaQSAR software platform. A comparison of classification-based validation metrics such as accuracy, sensitivity, specificity, and Matthews correlation coefficient (MCC) (see Table 33.4) for the test set (external validation) shows that the models developed with MetaQSAR outperformed the reported models. It is important to note that internal validation metrics were not provided for the models developed using QSPRmodeler; therefore, while internal validation metrics for MetaQSAR models are reported, a direct comparison was not possible.

4.4 Case study 4: Classification-based QSAR modeling for predicting toxicity of pesticides toward earthworms

In our final case study, we replicated a classification-based QSAR model originally developed to predict the toxicity of pesticides toward earthworms. Using the dataset of 163 diverse chemicals collected from the EFSA OpenFoodTox database, the authors of this study developed an LDA model to predict and characterize pesticide toxicity toward earthworm (*Eisenia foetida*). The toxicity data were expressed as LC₅₀ (mg/kg), and following OECD guidelines for “Earthworm Acute Toxicity Tests” (OECD, 1984), the authors had classified the compounds with LC₅₀ values below 1000 mg/kg as *toxic*, and those above 1000 mg/kg as *non-toxic*. We employed the same dataset and classification threshold for model development using the MetaQSAR platform. A comparison of classification-based validation metrics (see Table 33.4) for both internal and external validations indicates that the models developed using MetaQSAR performed better than the reported models.

Overall, across all case studies, the models generated by MetaQSAR showed performance that was comparable to, or in some cases exceeding, that reported in the literature.

5 Conclusion

In recent years, the development and adoption of automated QSAR platforms have significantly transformed the landscape of computational modeling in drug discovery, regulatory toxicology, and relevant fields. These platforms offer streamlined, reproducible, and scalable workflows for QSAR model development, enabling researchers to generate predictive models with minimal manual intervention while ensuring consistency with established best practices, such as those outlined by the OECD.

Already known automated QSAR tools such as AutoQSAR, DeepAutoQSAR, OCHEM, Open OCHEM, QSAR Framework (KNIME), QSAR Workbench using Pipeline Pilot, QSARtuna, QSPRmodeler, and AutoWeka have contributed substantially to improving accessibility and reliability in model development. They integrate various components of the QSAR workflow—data curation, descriptor calculation, feature selection, algorithm training, validation, and applicability domain assessment—into cohesive, often user-friendly systems. These platforms not only reduce the time and expertise required for model generation but also enhance reproducibility and transparency, which are critical for regulatory acceptance.

In this context, the **MetaQSAR** software platform stands out as an innovative and versatile addition to the suite of automated QSAR tools. This standalone tool is freely available for download, subject to user registration, at <https://chemopredictionsuite.com/MetaQSAR>. The MetaQSAR software platform offers a fully automated and user-friendly environment for the development of high-quality predictive models. Designed to accommodate both novice and expert users, the platform provides two flexible operational modes Automatic QSAR and Manual QSAR enabling users to choose between simplicity and customization based on their project needs.

Through streamlined workflows, MetaQSAR supports the entire modeling pipeline, from data preprocessing and descriptor calculation to model development, validation, and applicability domain assessment. The software adheres to OECD principles, ensuring that models are robust, interpretable, and regulatory-compliant. Its support for various machine learning algorithms, multiple dataset division strategies, and comprehensive validation metrics makes it a versatile tool for both regression and classification tasks. Notably, MetaQSAR simplifies the preparation of input files and generates a structured set of outputs categorized by workflow stages, enhancing transparency and reproducibility.

The comparative case studies presented in this chapter clearly demonstrate the efficacy and reliability of MetaQSAR in delivering models that perform on par with the models developed by QSAR experts or through other automated

platforms. Thus, MetaQSAR, in line with other automated QSAR platforms, serves as an invaluable resource for researchers, regulatory scientists, and industry professionals engaged in environmental risk assessment and cheminformatics-driven research.

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